# EXPLORATORY DATA ANALYSIS

## Telco Customer Churn Dataset

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*Dataset: IBM Telco Customer Churn (WA_Fn-UseC_-Telco-Customer-Churn.csv)*

*Total Records: 7,032 (after cleaning) | Features: 21*

*Date: April 2026*

| 7,032 | 26.5% | $548K | 21 |
|---|---|---|---|
| Customers Analyzed | Overall Churn Rate | Annual Revenue at Risk | Features Explored |

# 1. Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) of the IBM Telco Customer Churn dataset. The analysis covers 7,032 customers across 21 demographic, service, and billing features. The primary objective is to identify the key drivers of customer churn and quantify the associated financial risk.

The overall churn rate stands at 26.5%. However, specific customer segments exhibit dramatically higher churn, with the highest-risk cohort (Month-to-Month contract + Fiber Optic + Electronic Check) churning at over 53%. This report details each segment, the behavioral patterns observed, and strategic recommendations for retention.

## 1.1 Top 5 Critical Findings

- Fiber Optic customers churn at 41.9% vs. 7.4% for customers with no internet — a 5.6x gap.
- Senior citizens churn at 41.7%, nearly double the 23.7% rate for non-seniors.
- Customers with only 1 service have the highest churn at 45.8%; customers with 6 services churn at only 5.3%.
- Month-to-Month + Electronic Check is the most dangerous combination at 53.7% churn.
- $547,999 in annual recurring revenue is at immediate risk from 518 high-risk active customers.

# 2. Dataset Overview

## 2.1 Data Characteristics

The dataset originates from IBM Sample Data and contains information on a US telecom provider's customers. Each row represents one customer, with columns spanning demographics, service subscriptions, billing details, and churn status.

| Category | Detail | Value |
|---|---|---|
| Total Rows (raw) | Before cleaning | 7,043 |
| Rows removed | TotalCharges conversion errors | 11 |
| Final Dataset | After null removal | **7,032** |
| Features | Total columns | 21 |
| Target Variable | Churn (Yes / No) | **26.5% Yes** |
| Missing Values | After TotalCharges fix | None |

## 2.2 Data Cleaning Steps

- TotalCharges column was stored as a string and converted to numeric. 11 records with invalid (blank) values were dropped.
- SeniorCitizen was stored as 0/1 integer rather than Yes/No string — treated as binary.
- "No internet service" values in service columns were normalized to "No" to prevent dummy variable explosion.

# 3. Demographic Analysis

## 3.1 Gender Distribution

The dataset is nearly balanced by gender. Male customers account for 50.5% of the dataset and Female customers 49.5%, indicating no significant gender skew.

| Gender | Count | Churn Rate (%) |
|---|---|---|
| Male | 3,549 (50.5%) | 26.2% |
| Female | 3,483 (49.5%) | 27.0% |

*Gender is not a meaningful predictor of churn — the difference between male (26.2%) and female (27.0%) churn is negligible. Monthly charges are also nearly identical ($64.39 vs $65.22).*

## 3.2 Senior Citizen Behavioral Patterns

Although senior citizens represent only 16.2% of the customer base, they exhibit substantially different behavioral characteristics — particularly regarding churn and spending.

| Segment | Count (%) | Avg Monthly Charge | Churn Rate |
|---|---|---|---|
| Non-Senior (0) | 5,890 (83.8%) | $61.89 | 23.7% |
| Senior Citizen (1) | 1,142 (16.2%) | $79.82 | **41.7%** |

- Senior citizens pay approximately 29% more per month ($79.82 vs $61.89), likely due to premium packages.
- Despite paying more, seniors churn at nearly twice the rate of younger customers (41.7% vs 23.7%).
- This represents a critical business paradox — the highest-paying segment is also the most at-risk.

**Strategic Implication: Senior-specific loyalty programs, simplified billing, and proactive support channels should be prioritized to retain this high-value segment.**

## 3.3 Safety Net Analysis — Partner & Dependents

Family structure has a significant anchoring effect on customer loyalty. Customers with partners stay nearly twice as long and churn far less than solo customers.

| Household | Has Dependents | Churn Rate | Avg Tenure (Mo.) | Risk Level |
|---|---|---|---|---|
| No Partner | 9.9% | **33.0%** | ~23.4 | **HIGH** |
| Has Partner | 51.3% | **19.7%** | ~42.1 | **STABLE** |

- Solo customers (no partner, no dependents) represent the highest-risk profile — more mobile, more likely to respond to competitor promotions.
- Tech Support significantly increases tenure for solo customers, suggesting that adding services can substitute for household anchoring.

# 4. Service Stickiness Analysis

This section examines how the breadth of service subscriptions (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies) correlates with customer retention. A "Service Count" score (0–6) was engineered for each customer.

## 4.1 The Loyalty Cliff

| Services | 0 | 1 | 2 | 3 | 4–6 |
|---|---|---|---|---|---|
| **Churn Rate** | 21.5% | **45.8%** ⚠ | 35.8% | 27.4% | 5–12% |

- Customers with exactly 1 service exhibit the highest churn (45.8%) — they are in the ecosystem but not yet integrated.
- Customers with 0 services (phone-only) churn at 21.5%, likely because they have lower price sensitivity.
- Each additional service acts as a switching cost. At 5–6 services, churn drops to just 5–12%.

**Recommendation: Cross-sell strategies should target 1-service customers urgently. Bundle promotions that move customers from 1 to 3+ services will have the highest ROI for churn reduction.**

# 5. Internet Service Analysis

## 5.1 Churn by Internet Technology

The type of internet service is one of the strongest predictors of churn in the dataset. Fiber Optic customers, while generating the most revenue, are also the most likely to leave.

| Internet Service | Churn Rate | Avg Monthly Charge | Total Customers |
|---|---|---|---|
| No Internet | 7.4% | $21.08 | 1,520 |
| DSL | 19.0% | $58.09 | 2,416 |
| Fiber Optic | **41.9%** | $91.50 | 3,096 |

- Fiber Optic customers generate the most revenue ($91.50/month avg.) but churn at 41.9% — far above the 26.5% company average.
- These customers have higher expectations for service quality and are more likely to switch if dissatisfied.
- Among Fiber Optic customers, having multiple lines reduces churn slightly (39.6% vs. 45.8% for single-line).
- No-internet customers represent a very stable base at 7.4% churn, likely because their low monthly costs ($21) eliminate the cost-benefit case for switching.

**Strategic Implication: Fiber Optic customers require dedicated quality assurance, proactive outreach, and premium support to reduce churn from this critical revenue segment.**

# 6. Billing & Payment Analysis

## 6.1 Churn by Payment Method

Payment method reveals customer behavioral patterns beyond just billing preferences. Electronic Check users exhibit dramatically higher churn — not necessarily because of the payment method itself, but because of what it signals about the customer.

| Payment Method | Churn Rate | Avg Monthly Charge | Customer Count |
|---|---|---|---|
| Credit Card (Automatic) | 15.3% | $66.52 | 1,521 |
| Bank Transfer (Automatic) | 16.7% | $67.21 | 1,542 |
| Mailed Check | 19.2% | $43.96 | 1,604 |
| Electronic Check | **45.3%** | $76.26 | 2,365 |

- Electronic Check users churn at 45.3% — nearly 3x the rate of automatic payment users (~15–17%).
- Electronic check creates a "monthly re-evaluation cycle" — every payment is an active decision, not a passive auto-debit, exposing the company to intentional cancellation.
- Automatic payment methods (credit card and bank transfer) are strongly correlated with retention, likely reflecting higher-commitment, digitally-engaged customers.

## 6.2 Churn Risk Matrix — Contract × Payment Method

Combining contract type and payment method reveals the full spectrum of churn risk across the customer base. The heatmap below summarizes churn percentages across these two dimensions.

| Contract | Bank Transfer (Auto) | Credit Card (Auto) | Electronic Check | Mailed Check |
|---|---|---|---|---|
| **Month-to-Month** | 34.1% | 32.8% | **53.7%** | 31.6% |
| One Year | 9.7% | 10.3% | 18.4% | 6.9% |
| Two Year | 3.4% | 2.2% | 7.7% | 0.8% |

- Month-to-Month + Electronic Check is the most dangerous combination: 53.7% churn rate.
- Long-term contracts are a powerful retention tool regardless of payment method — Two Year contracts keep churn below 8% even for electronic check users.
- Two Year + Mailed Check is the most stable combination at just 0.8% churn.

# 7. Financial Risk Analysis

This section quantifies the direct revenue at risk from the highest-churn customer segment that has not yet churned — providing a financial priority score for retention campaigns.

## 7.1 High-Risk Segment Definition

The "Danger Zone" is defined by the intersection of the four strongest churn indicators identified in this analysis:

- Contract Type: Month-to-Month (no long-term commitment)
- Internet Service: Fiber Optic (highest churn internet segment)
- Payment Method: Electronic Check (highest payment-related churn)
- Churn Status: Currently Active (customers who have not yet churned but exhibit high-risk behavior)

| Metric | Value |
|---|---|
| High-Risk Customers Remaining | **518 customers** |
| Monthly Revenue at Risk (MRR) | **$45,666.60** |
| Annual Revenue at Risk (ARR = MRR × 12) | **$547,999.20** |
| Total Company Monthly Revenue | **$316,530.15** |
| Revenue in Danger Zone (%) | **14.43%** |

Nearly 14.4% of total monthly revenue is exposed to high churn risk from just 518 customers — 7.4% of the active base. This makes the Danger Zone segment a disproportionately high priority for immediate retention interventions.

# 8. Tenure & Correlation Analysis

## 8.1 Churn Rate by Tenure Group × Contract

Customer tenure is a strong inverse predictor of churn. New customers are dramatically more likely to leave, and the risk decreases substantially over time — particularly for Month-to-Month contracts.

| Tenure Group | Month-to-Month | One Year | Two Year |
|---|---|---|---|
| 0–1 Year | **51.4%** | 10.6% | 0.0% |
| 1–2 Years | 37.7% | 8.1% | 0.0% |
| 2–3 Years | 32.5% | 8.0% | 2.1% |
| 3–4 Years | 33.5% | 13.1% | 2.2% |
| 4–5 Years | 27.8% | 13.7% | 4.0% |
| 5+ Years | 22.2% | 12.1% | 3.1% |

- The first year is the highest-risk period: 51.4% of new Month-to-Month customers leave within 12 months.
- Two-Year contract customers in Year 1 churn at 0% — demonstrating the immense retention value of long-term commitments.
- KDE analysis confirms a bimodal pattern: churners cluster heavily in the 0–20 month range, while retained customers are broadly distributed through 60+ months.

## 8.2 Numerical Feature Correlations

Correlation analysis among tenure, MonthlyCharges, and TotalCharges reveals important structural relationships:

| Feature Pair | Correlation | Interpretation | Implication |
|---|---|---|---|
| Tenure ↔ TotalCharges | **0.826** | Very Strong | Longer customers = more lifetime value |
| MonthlyCharges ↔ TotalCharges | 0.651 | Moderate | High spenders accumulate faster |
| Tenure ↔ MonthlyCharges | 0.247 | Weak | Longer tenure does not predict higher bills |

# 9. Strategic Recommendations

Based on the EDA findings, the following prioritized recommendations are proposed to reduce churn and protect revenue:

## 9.1 Immediate Priority — Danger Zone Intervention

- Identify the 518 active customers in the Month-to-Month + Fiber Optic + Electronic Check segment and launch a targeted retention campaign.
- Offer contract upgrade incentives (e.g., discounted One-Year contracts) to reduce $547,999 in annual revenue exposure.
- Automate a "churn risk flag" in CRM systems for customers matching this profile.

## 9.2 High Priority — Senior Citizen Loyalty Program

- Design a Senior-Specific loyalty program with simplified billing, priority support, and price-lock guarantees.
- Assign dedicated account representatives to senior customers in the highest monthly charge tier.
- Monitor senior churn monthly — a reduction from 41.7% to 30% would retain approximately 133 senior customers annually.

## 9.3 Medium Priority — Service Cross-Sell Campaign

- Target customers with exactly 1 service subscription (45.8% churn risk) with bundled upgrade offers.
- Promote TechSupport, OnlineSecurity, and OnlineBackup bundles — these services consistently lower churn across all segments.
- Goal: Move at-risk single-service customers to 3+ services to reduce their churn probability below 28%.

## 9.4 Long-Term — Payment Method Migration

- Incentivize Electronic Check users to migrate to automatic payment methods through discounts or reward points.
- Implement a "paperless billing with auto-pay" promotional bundle targeted at high-churn payment segments.
- Goal: Reduce the share of Electronic Check payers from 33.6% to below 20% within 12 months.

# 10. Feature Engineering for Modeling

As a foundation for predictive modeling, the following engineered features were derived from the EDA insights:

| Feature | Definition | Insight |
|---|---|---|
| Total_Services | Count of add-on services subscribed (0–6) | Higher count strongly predicts retention |
| Is_Solo | 1 if no Partner AND no Dependents | Solo customers are highest churn risk |
| Is_Auto_Pay | 1 if automatic payment method | Auto-pay correlates with ~3x lower churn |
| Short_Contract | 1 if Month-to-Month contract | Strongest individual churn predictor |
| Charge_Spike | MonthlyCharges − (TotalCharges / tenure) | Detects recent price increases above history |

*End of Report*

*Telco Customer Churn EDA — April 2026*